

Paper 25.25 - Energetic Traversal Toolkit Supplement

CQE-paper-25.25

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-25.25.md

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Purpose

This supplement shows how to reproduce Paper 25's traversal receipt. The proof is in Paper 25 and its formal verifier; this file holds the tool route and the analog route.

Digital Route

Run:

```
`python production/formal-papers/CQE-paper-25/verify_energetic_traversal.py`
```

The expected status is ``pass_with_open_obligations``. The verifier checks ``NSLTerm`` arithmetic, the four-row transport-obligation spine, the ``2 + 6`` VOA default analog cost, additive path totals, one closed normalized traversal, and one open uncalibrated traversal.

Analog Route

Draw each transport as a row with source, target, and ``(L, C, R)``. Write the four NSL fields beside it: conservation residue, information residue, irreversible cost, and absorption. Add the first three values, subtract absorption, and mark the row closed only when the result is less than or equal to zero.

To expose the path, thread the rows in order. The path total is the sum of the row totals. A negative or zero path may close only when no row is marked uncalibrated. Any positive row or uncalibrated row is carried forward as an obligation.

Hidden-Guess Diagnostic

Training mode should hide the label until after the reviewer chooses one:

- closing NSL step,
- non-closing NSL step,
- closing normalized traversal,
- open traversal,
- uncalibrated physical-energy promotion,
- invalid least-action promotion,
- valid obligation carry.

The answer key distinguishes analog accounting closure from physical-energy closure.